

Statistical Modeling in R

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About Me



- **MSc in Statistics, Dal**
- **Data quality control analyst, Dal**
- **Statistical modeling specialist**
- **Workshop facilitator**
- **New ocean diver**

PART 1

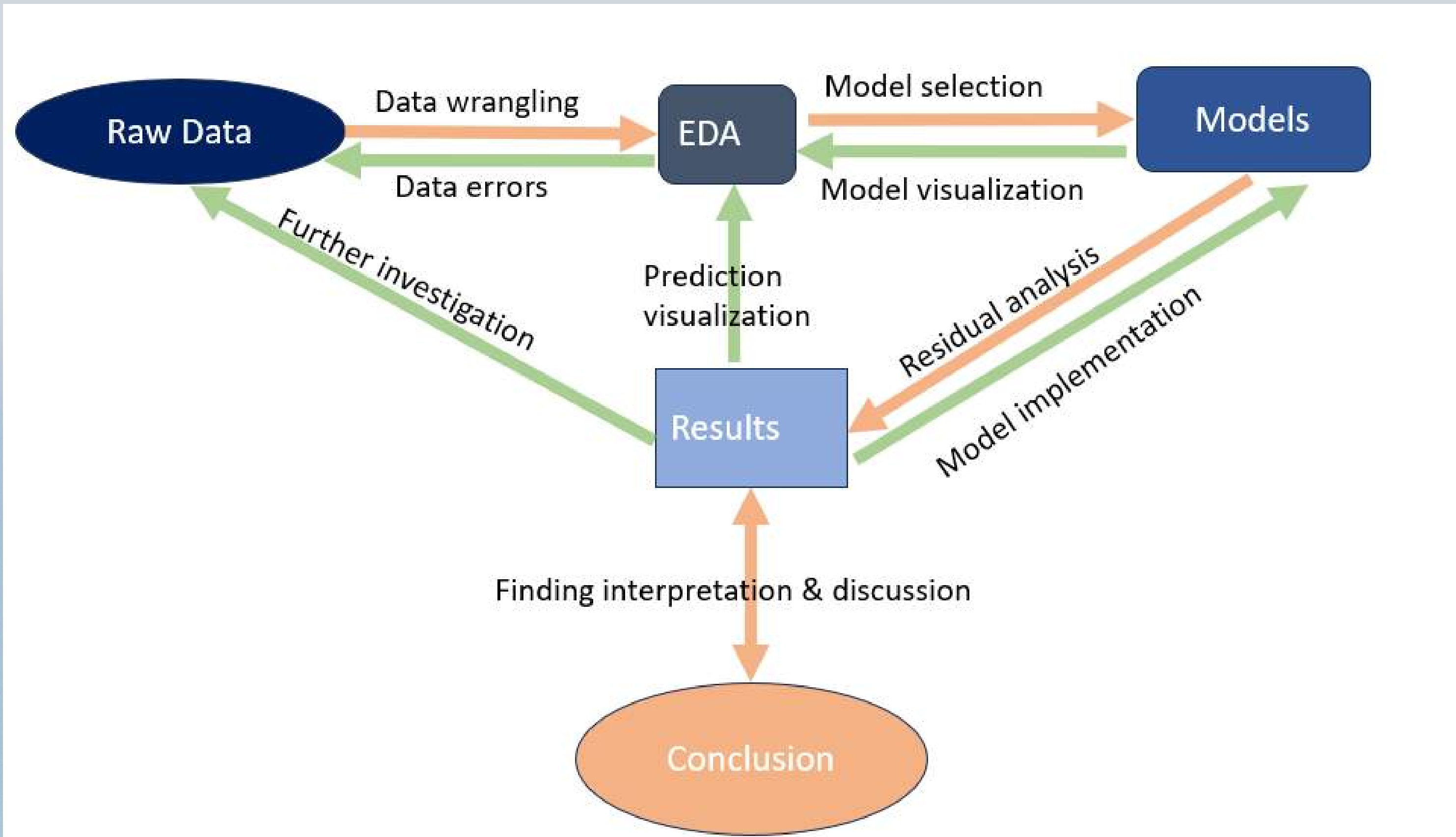
Foundation

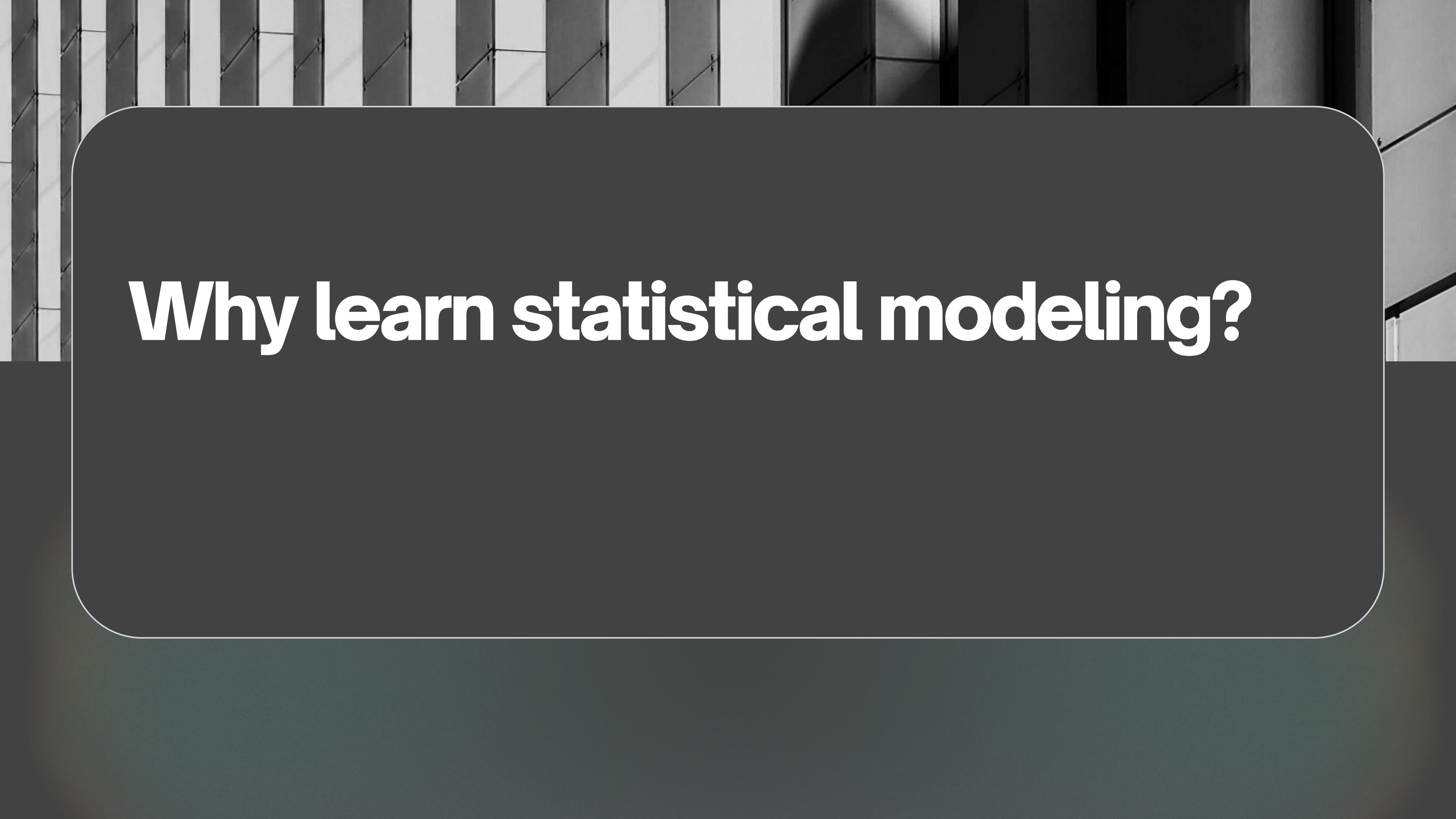
May 5th, 2026

Workshop Overview

- 01** Data Analysis Workflow
- 02** Common Statistical Models

Data Analysis Workflow





Why learn statistical modeling?

Advantages of Statistical Models

● INTERPRETABILITY

Statistical models often directly show how each variable influences the outcome(e.g., coefficients, significance).

👉 This makes them ideal for explaining results to stakeholders and supporting scientific conclusions.

● BETTER FOR DATA QUALITY & DIAGNOSTICS

Statistical models provide built-in tools (residuals, assumptions checks) to detect:

- data issues
- model misspecification

👉 This helps improve both data quality and model reliability.

● EFFICIENT WITH SMALLER & STRUCTURED DATA

Statistical models perform well even with:

- smaller datasets
- well-designed experiments

👉 Unlike many ML models, they don't require large amounts of data or heavy computation.

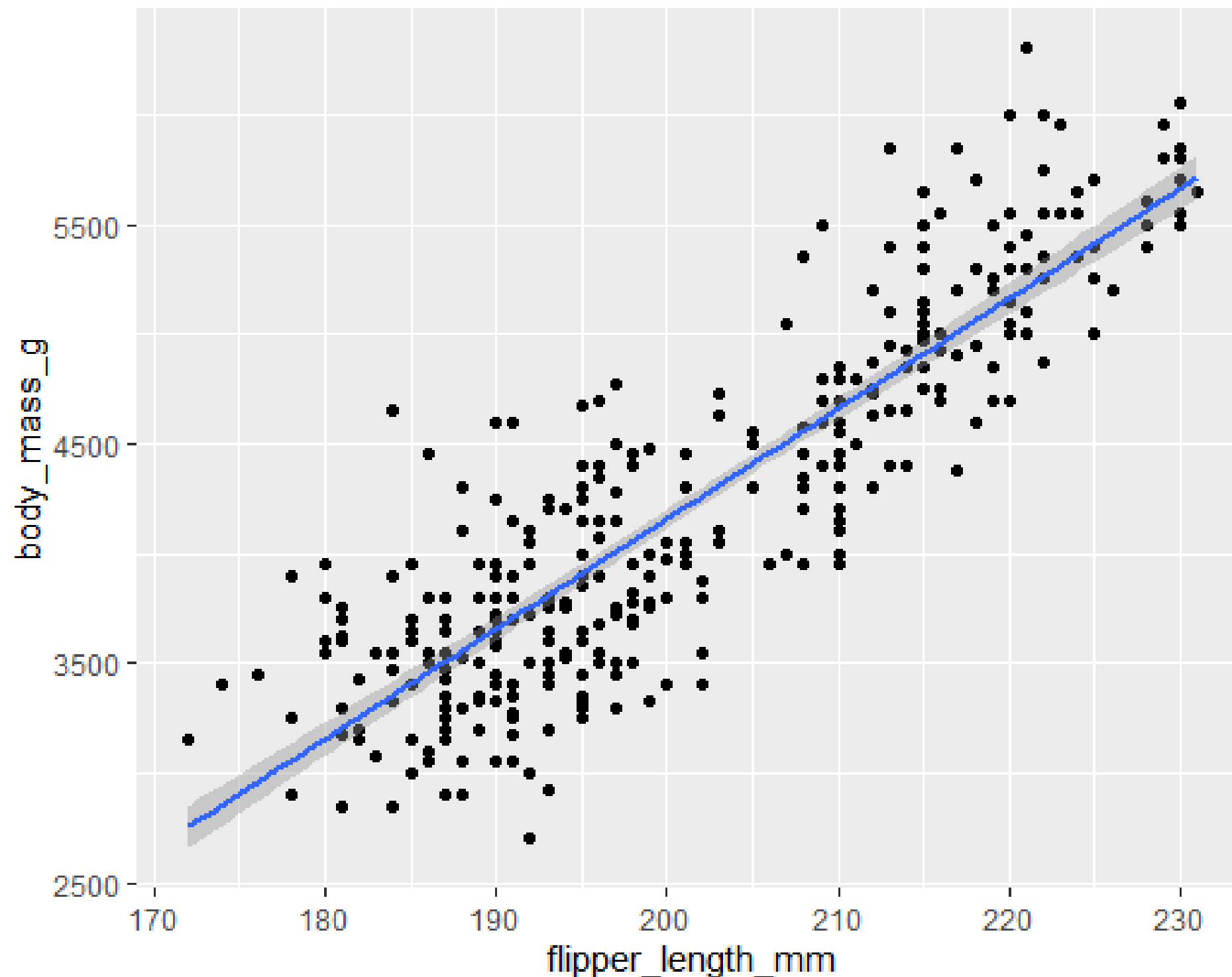
It's always good to know more tools in the data world! 😊

Common Statistical Models

- 01** Linear Model (LM)
- 02** Generalized Linear Model (GLM)
- 03** Generalized Additive Model (GAM)
- 04** Generalized Linear Mixed Model (GLMM)
- 05** Generalized Additive Mixed Model (GAMM)

Linear Model (LM)

$$y = X\beta + \varepsilon, y \sim N(\mu, \sigma^2), \beta = C$$



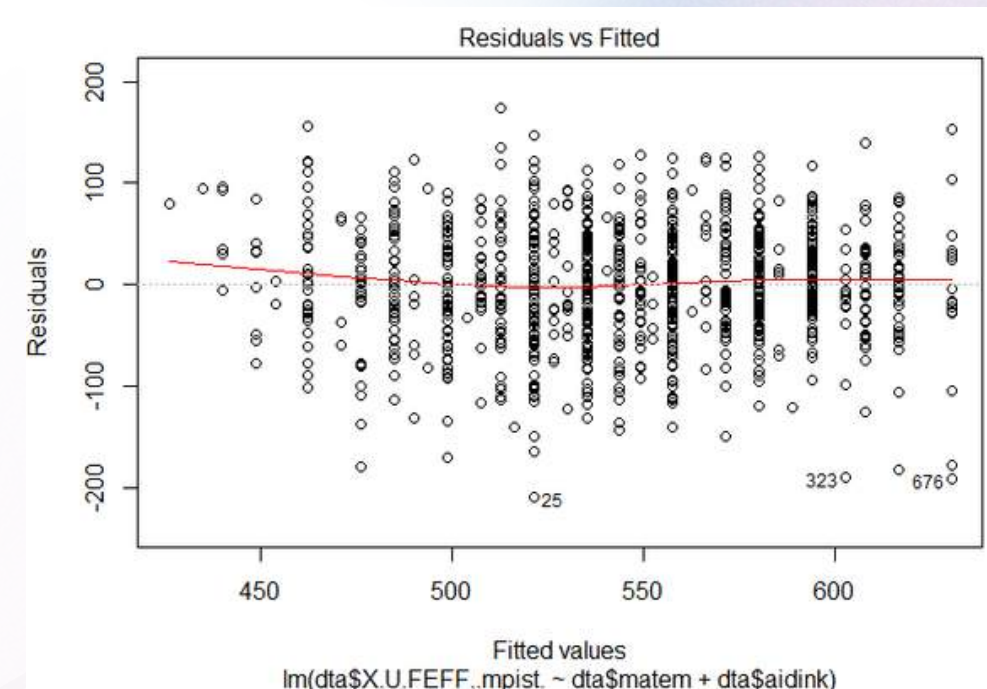
Key Assumptions

- **Linearity:** The relationship between input and output is linear.
- **Homogeneity of variance (Homoscedasticity):** The error variance is constant across predictors.
- **Independence:** Observations are independent.
- **Normality:** The response and residuals are normally distributed

LM Residual Analysis

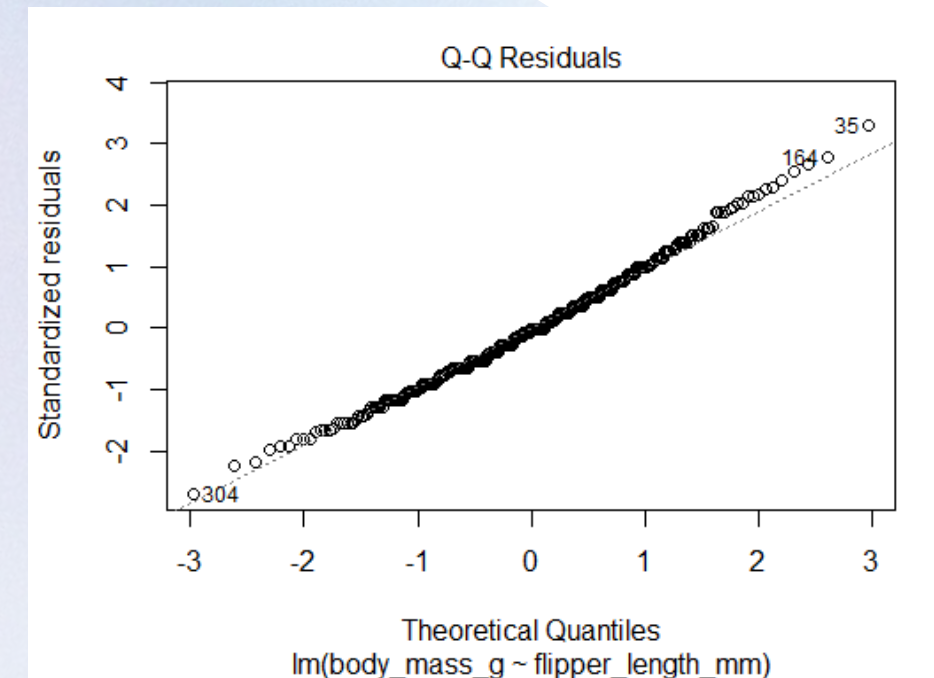
Linearity → Residuals vs Fitted

a flat horizontal line at zero with no distinct pattern is ideal.



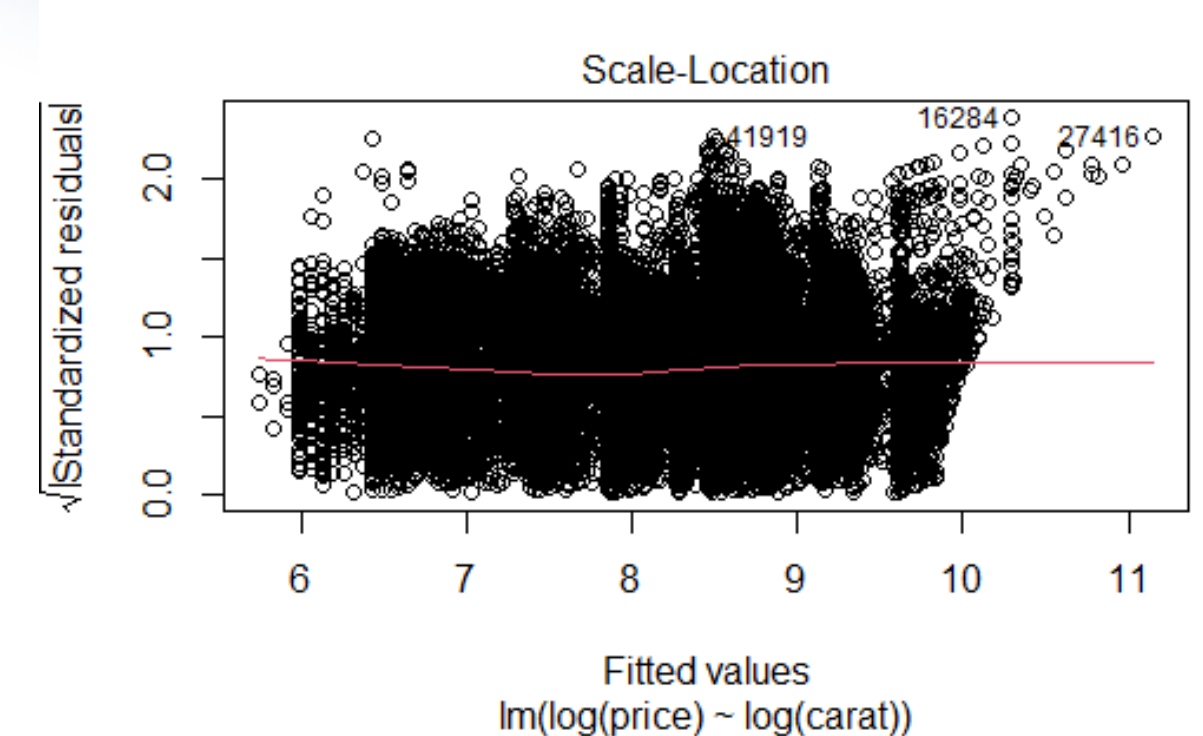
Normality → QQ plot

points should follow the diagonal dashed line.

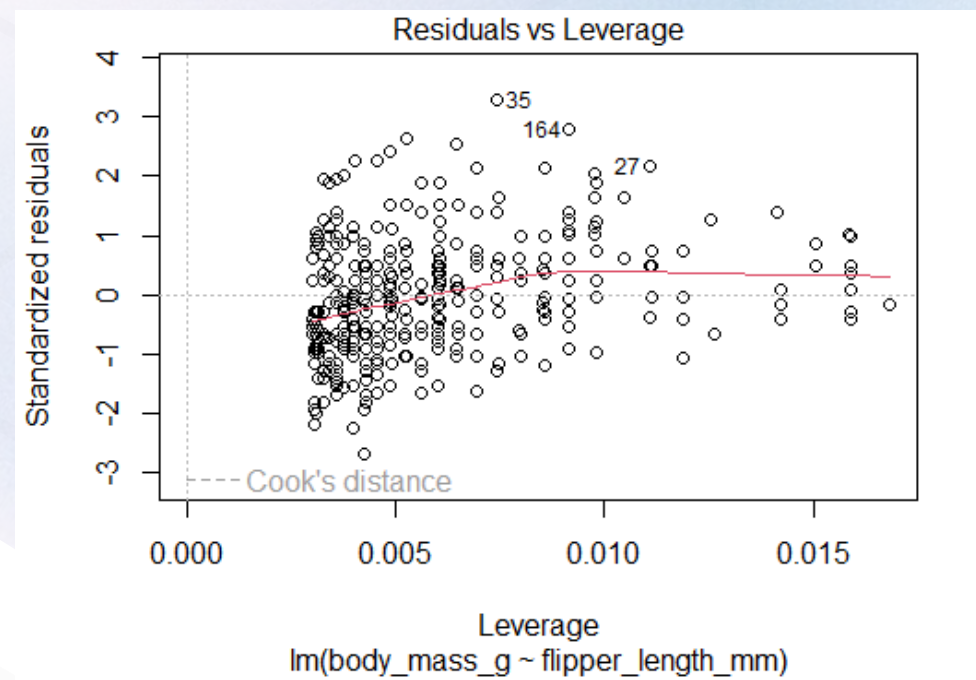


Homoscedasticity → Scale-Location

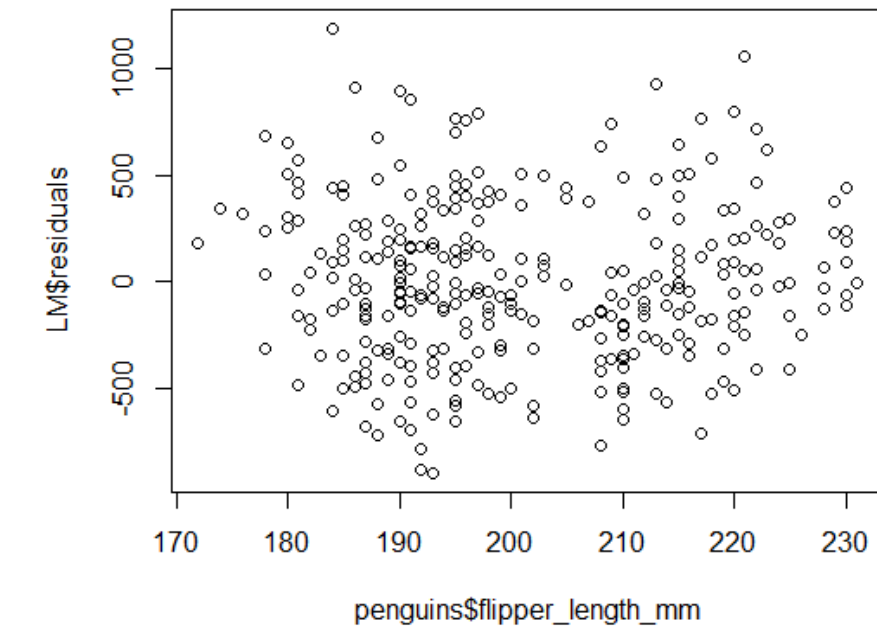
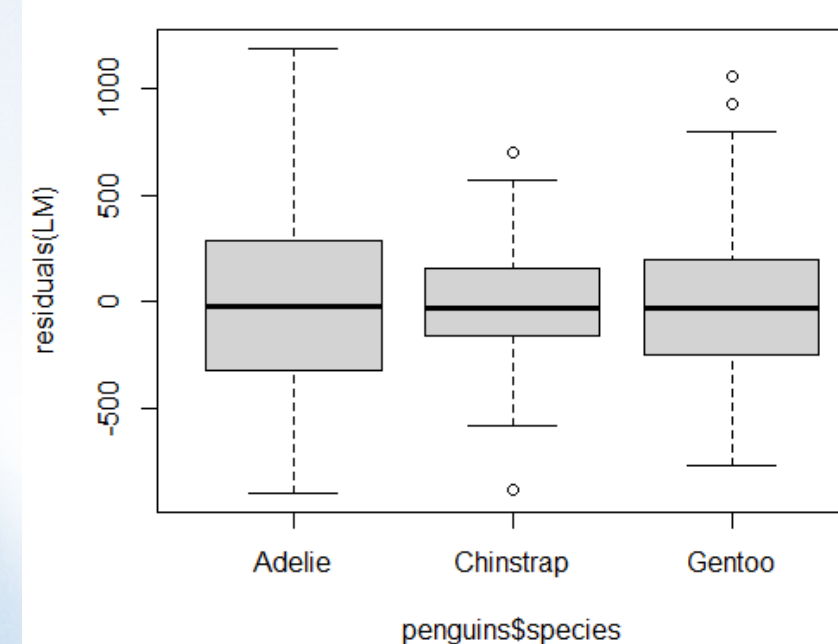
points should be spread equally around a horizontal line.



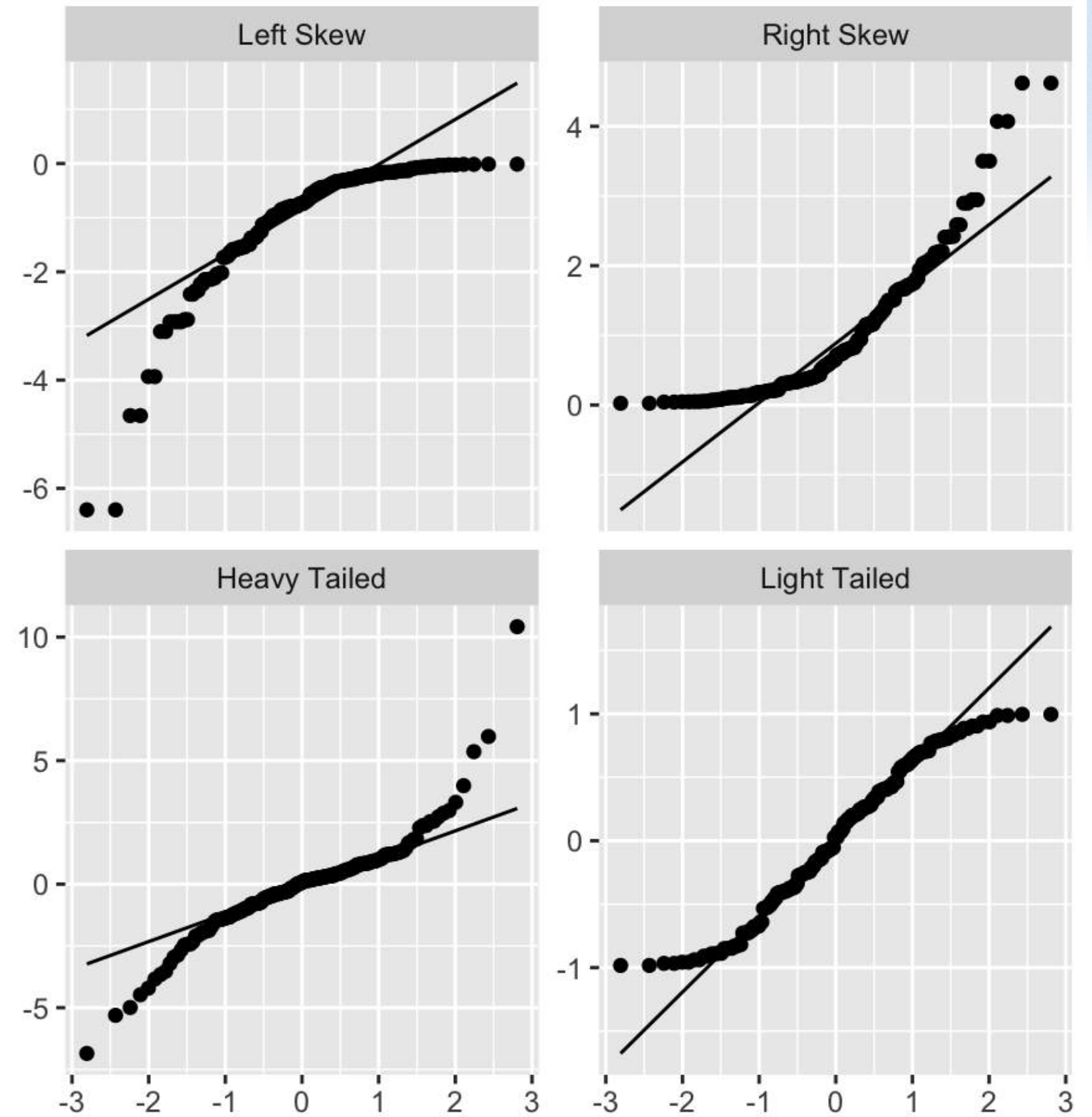
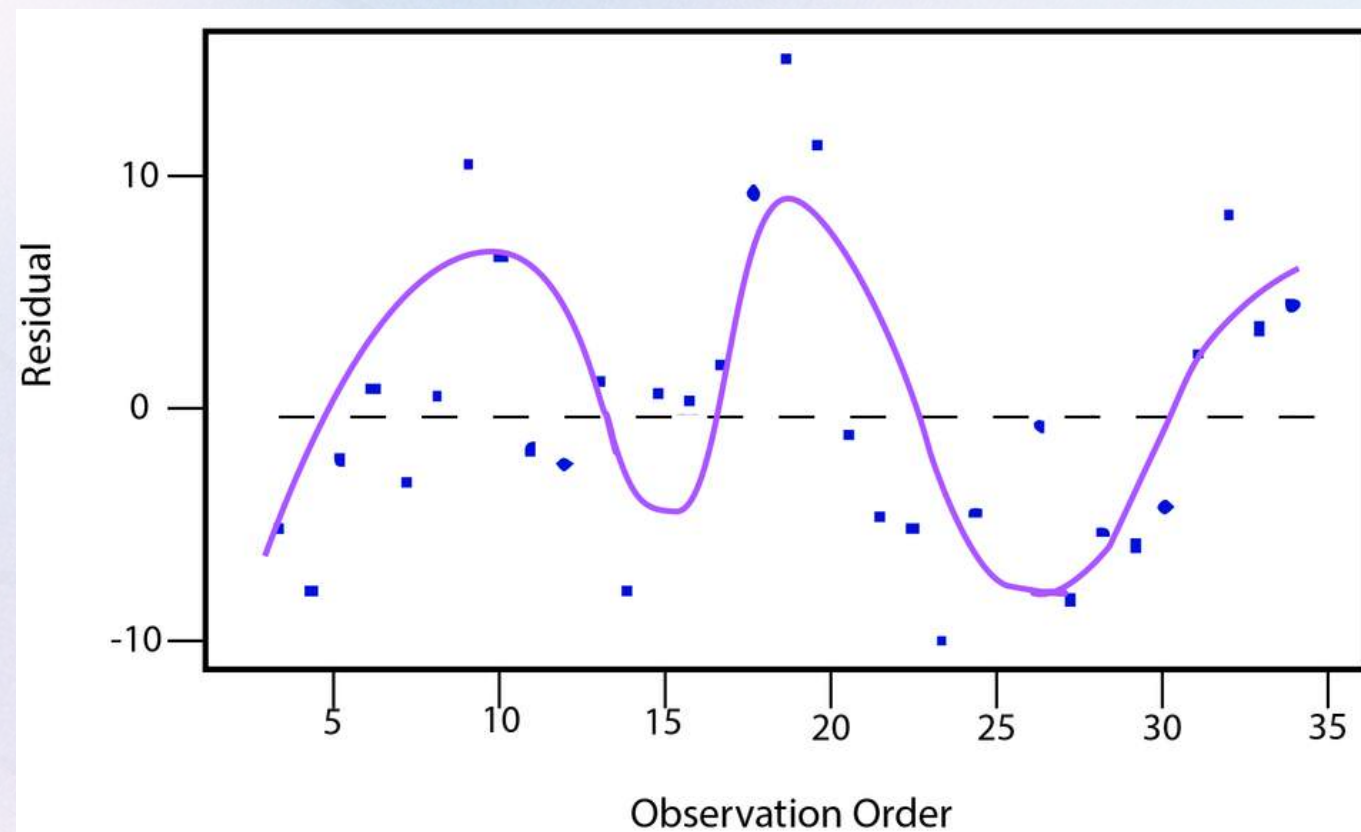
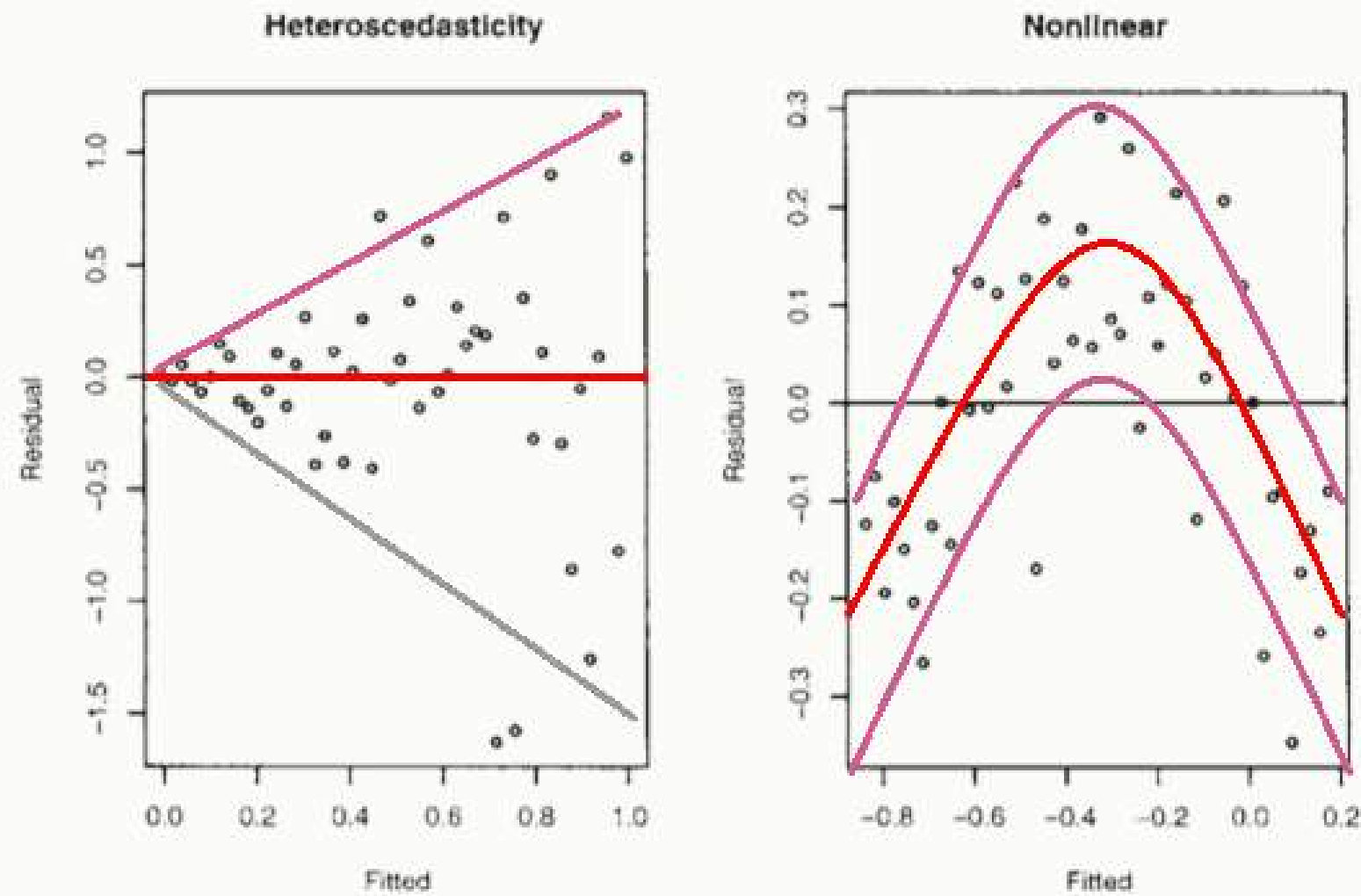
Influential observations → Residuals vs Leverage



Independence → Residuals vs Each explanatory variable



“Bad” Outputs



Generalized Linear Model (GLM)

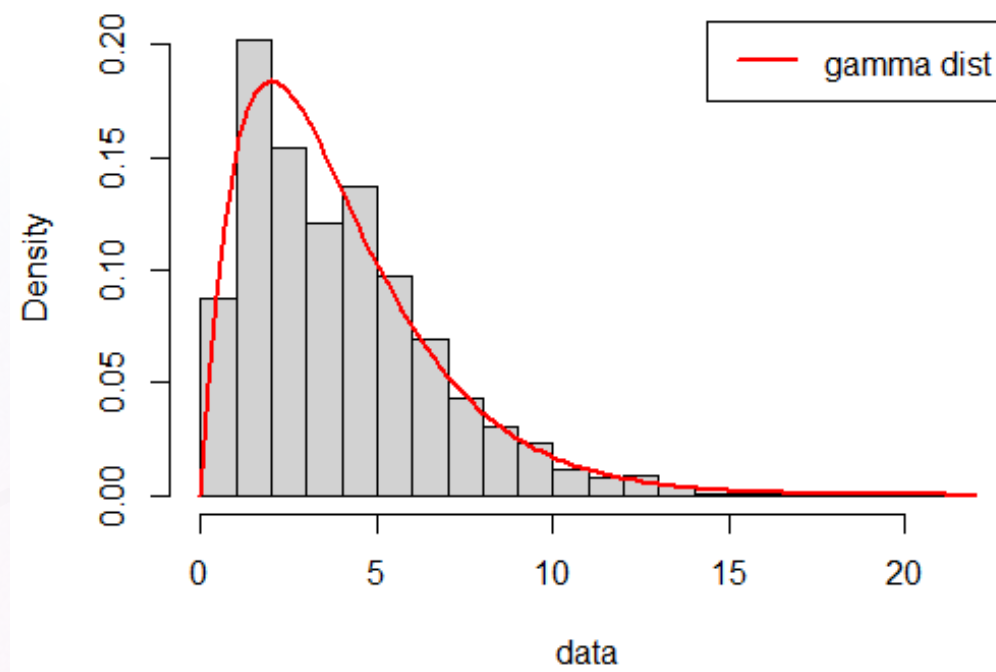
$$g(y) = X\beta + \varepsilon, y \sim E(\boldsymbol{\theta}), \beta = \mathbf{C}$$

Key Assumptions

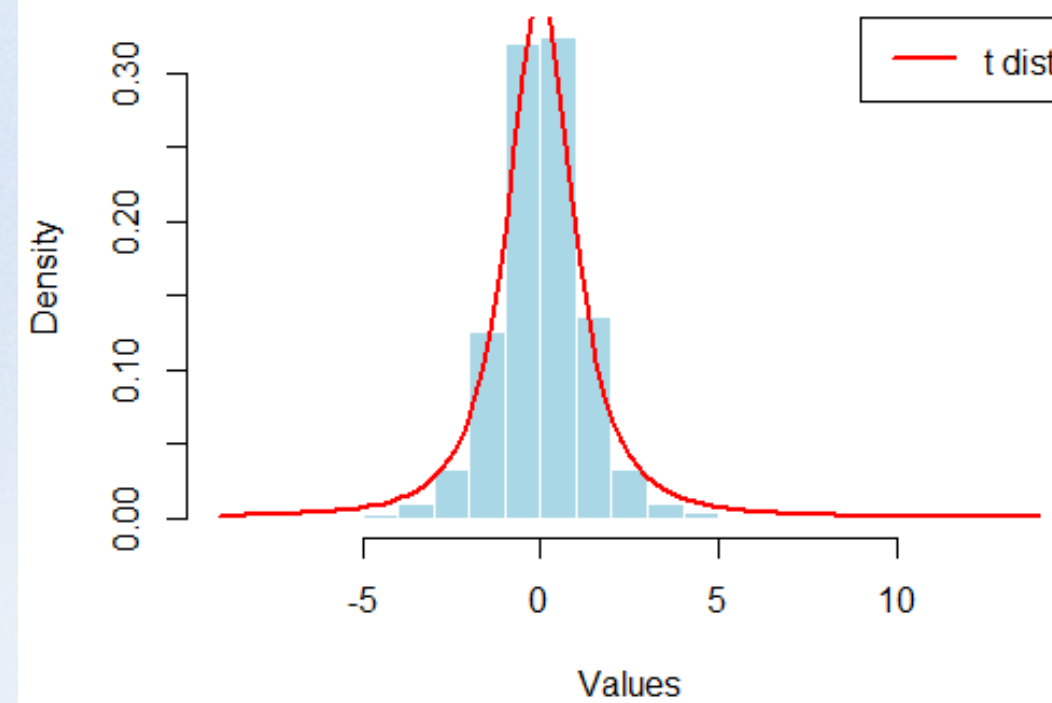
- **Correct Distribution:** The response variable follows a distribution from the exponential family (e.g., Binomial, Poisson, Gamma, Normal).
- **Linearity:** The transformation of the response $g(y)$ is linearly related to the explanatory variables βX
- **Independence:** Observations are independent.

Exponential family distributions

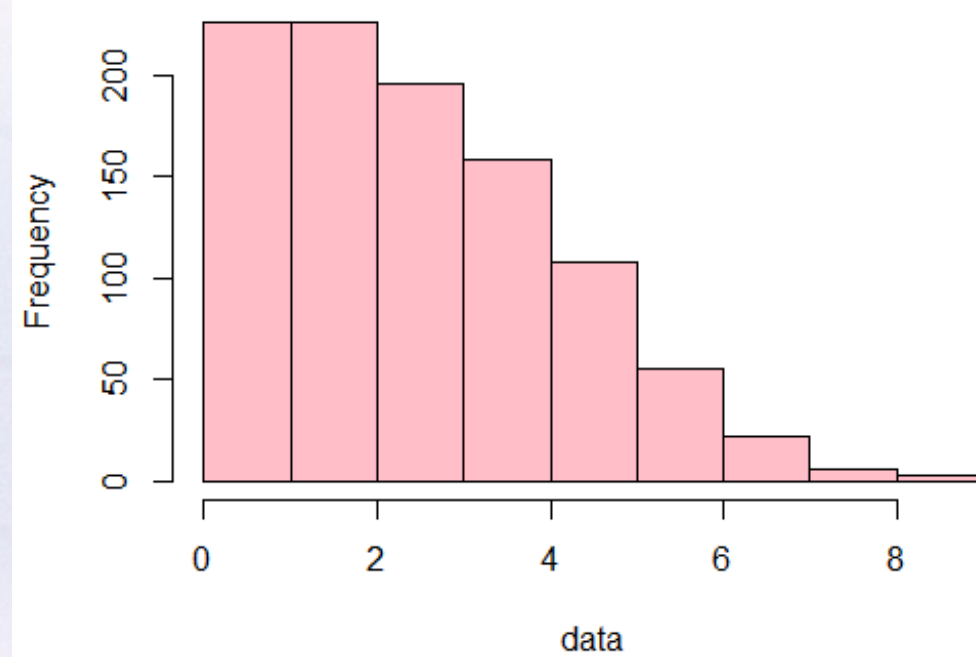
Histogram with Gamma Fit



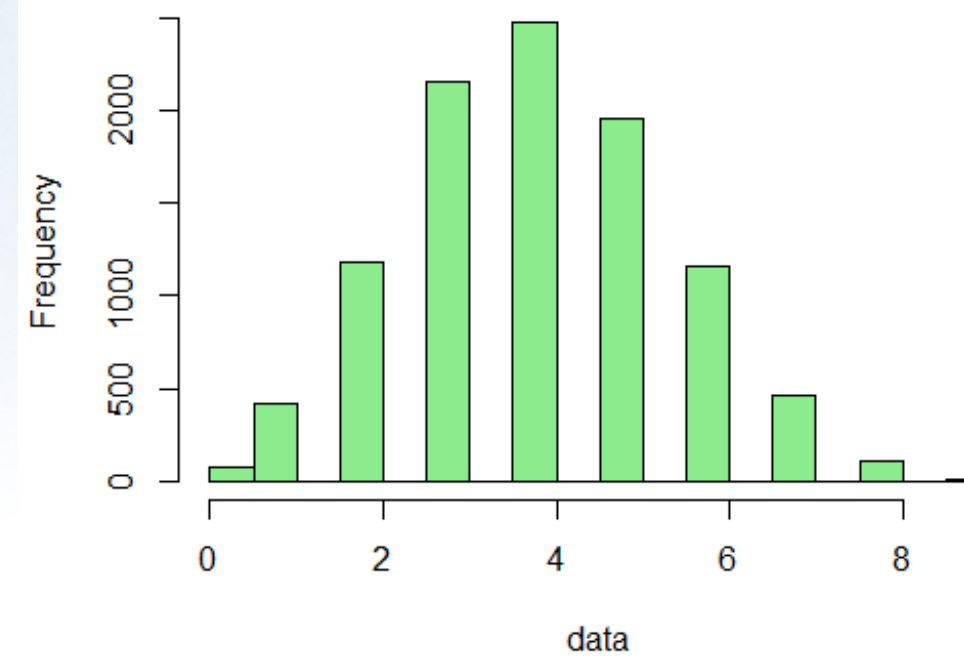
Histogram of T-Distribution



Poisson Distribution Histogram



Binomial Data Histogram



Distribution selection strategy

01

Identify the Response Type

- Continuous (numeric values)

☛ Examples: weight, temperature

Common choices:

Normal → symmetric data

Gamma → positive & skewed

t-distribution → heavy tails

Beta → bounded (0–1)

- Discrete (counts: 0, 1, 2, ...)

☛ Examples: number of events

Common choices:

Poisson → mean $\approx$ variance

Negative Binomial → overdispersion (variance > mean)

- Categorical (classes / labels)

1. Binary (2 categories)

☛ Yes/No, 0/1 → Logistic regression

2. Multiclass (unordered)

☛ North, South, East, West

Common choices:

Multinomial logistic

Random forest

02

Explore the Data (EDA)

- Plot the response (e.g., histogram)
- Look at shape: symmetric, skewed, bounded, counts

☛ Choose distributions that match the data pattern

03

Fit Candidate Models

- Try multiple reasonable distributions
- ☛ (e.g., Normal vs Gamma, Poisson vs Negative Binomial)

04

Compare & Diagnose Models

- Check residuals (patterns, variance, normality)
 - Evaluate model fit
- ☛ Select the model that best fits the data and assumptions

GLM Residual Analysis

LM vs GLM

- LM assumes normal responses (sepecial case in GLM)
- GLMs allow non-normal responses (e.g., Poisson, Binomial)
- ☛ Standard residuals may not behave well

Randomized Quantile Residuals (RQR)

How it works:

- Transform residuals to be approximately normal
- Works across different GLM distributions
- ☛ Makes QQ plots meaningful again
- Reference: Dunn & Smyth (1996)

What to Check:

- QQ plot → should follow straight line
- Residual vs fitted → no clear pattern
- Outliers → unusually large residuals

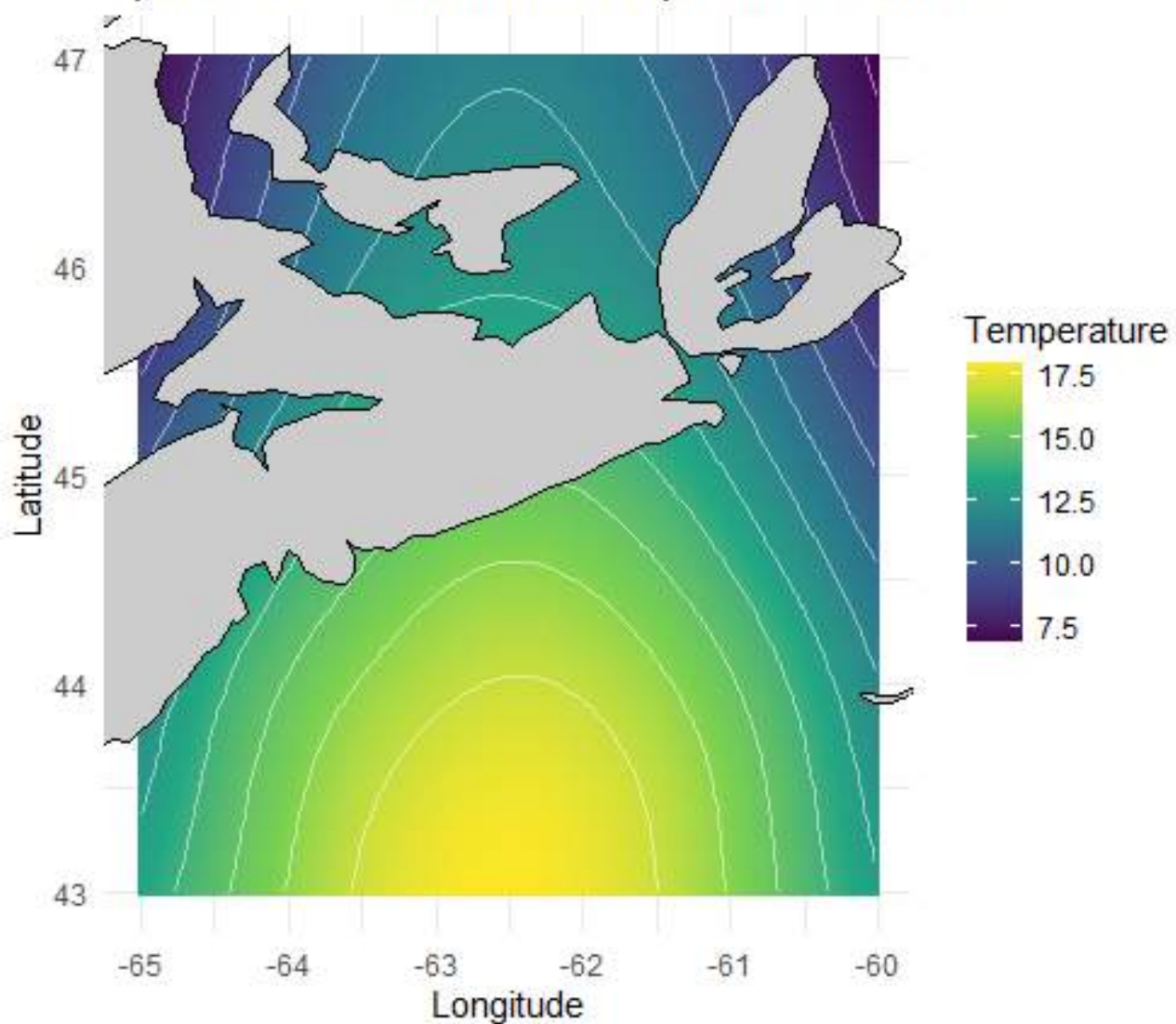
R Tools

- `statmod::qresid()` → compute RQR
- DHARMA → simulation-based residual diagnostics
 - ☛ Built-in plots
 - ☛ Works for many GLMs / GLMMs

Generalized Additive Model (GAM)

$$g(y) = a + f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n) + \varepsilon, y \sim E(\boldsymbol{\theta})$$

Spatial GAM: Simulated Temperature Surface

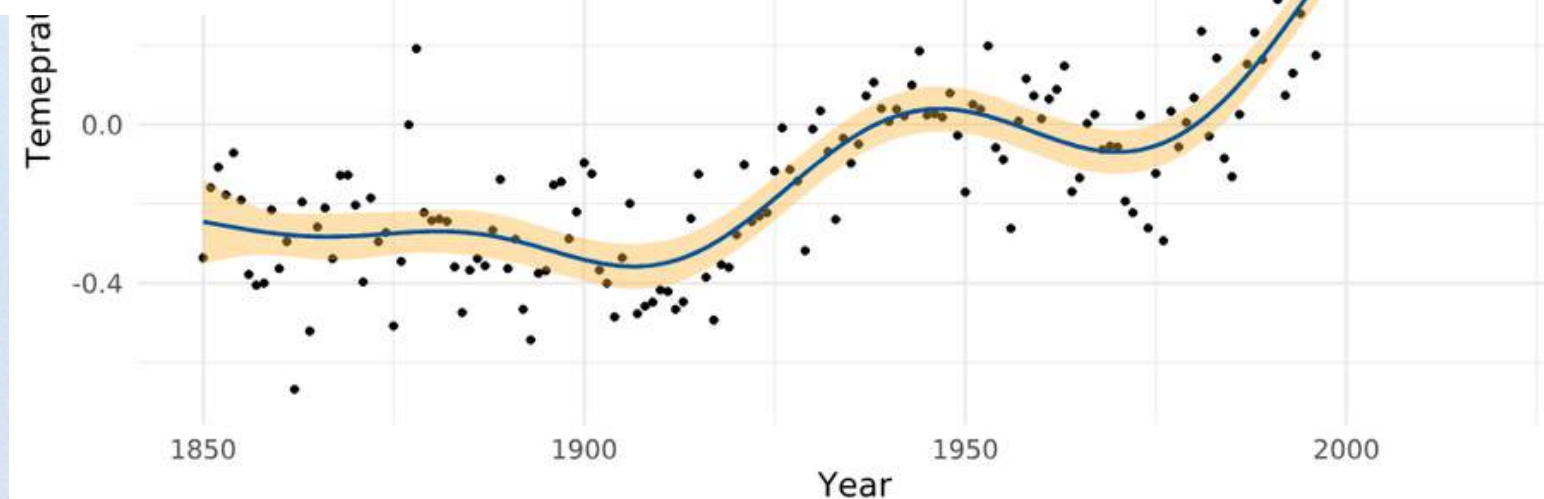


Key Assumptions

- **Correct Distribution**
- **Independence**

Key Characteristics

- **Non-linear Relationships:** GAMs use **smoothing functions**: automatically or manually detect model non-linear relationships.
- **Additive Structure:** The model relates the response variable to predictors through sum of smooth functions $f_j(x_j)$
- **Flexible Distributions:** Same as GLMs.
- **Interpretability:** Unlike "black box" models, GAMs permit **inspection of each smooth function** to understand how a specific predictor affects the outcome.



Smooth Functions

Function	When to Use	How to Use in R	Method (Idea)	Output Interpretation
s(x)	1D nonlinear effect	y ~ s(x)	Spline (basis expansion + penalty)	Smooth curve of x vs y
s(x, k=...)	Control flexibility	y ~ s(x, k=10)	More basis functions = more wiggle	Higher k → more flexible curve (knots and degree)
te(lon,lat)	Interaction (2D smooth)	y ~ te(lon, lat)	Tensor product smooth	Surface (lon & lat jointly affect y)
ti(x, z)	Interaction + main effects separate	y ~ s(x) + s(z) + ti(x,z)	ANOVA-style decomposition	Interaction effect only
t2(x, z)	Stable tensor smooth (advanced)	y ~ t2(x, z)	Alternative tensor basis	Similar to te() but more robust
s(x, bs="cs")	Shrinkage smooth	y ~ s(x, bs="cs")	Penalized spline with shrinkage	Can shrink to zero effect
s(x, bs="re")	Random effect	y ~ s(group, bs="re")	Random intercepts	Group-level variation

LM vs GLM vs GAM — Summary

Model	When to Use	Key Assumptions	R Package / Function	What to Check
LM	Continuous response, linear relationship	Linearity Normality Homogeneity Independence	stats::lm()	Residual vs fitted QQ plot Scale-Location Residuals vs each explanatory variable Residuals vs Leverage
GLM	Non-normal response (binary, counts)	Linearity Independence	stats::glm()	Use RQR to check everything above Check overdispersion for Poisson distributions
GAM	Nonlinear relationships	Independence	mgcv::gam()	Smooth plots RQR diagnostics Check k / wiggleness for overfitting

When Independence Assumption Breaks

Hospital Name	Patient ID	Treatment	Dosage	Hospitalization duration	Disease Type	Sex
QII	1	A	75 ug	1 month	Disease X	F
QII	2	A	70 ug	1 month	Disease X	F
QII	3	B	80 ug	1.5 month	Disease X	F
QII	4	B	40ug	2 month	Disease X	M
QII	5	C	50 ug	0.5 month	Disease X	M
IWK	6	A	75 ug	1 month	Disease X	F
IWK	7	A	70 ug	1 month	Disease X	F
IWK	8	C	80 ug	0.5 month	Disease X	F
IWK	9	B	40ug	2 month	Disease X	M
IWK	10	A	50 ug	3 month	Disease X	M

LM / GLM / GAM assume:

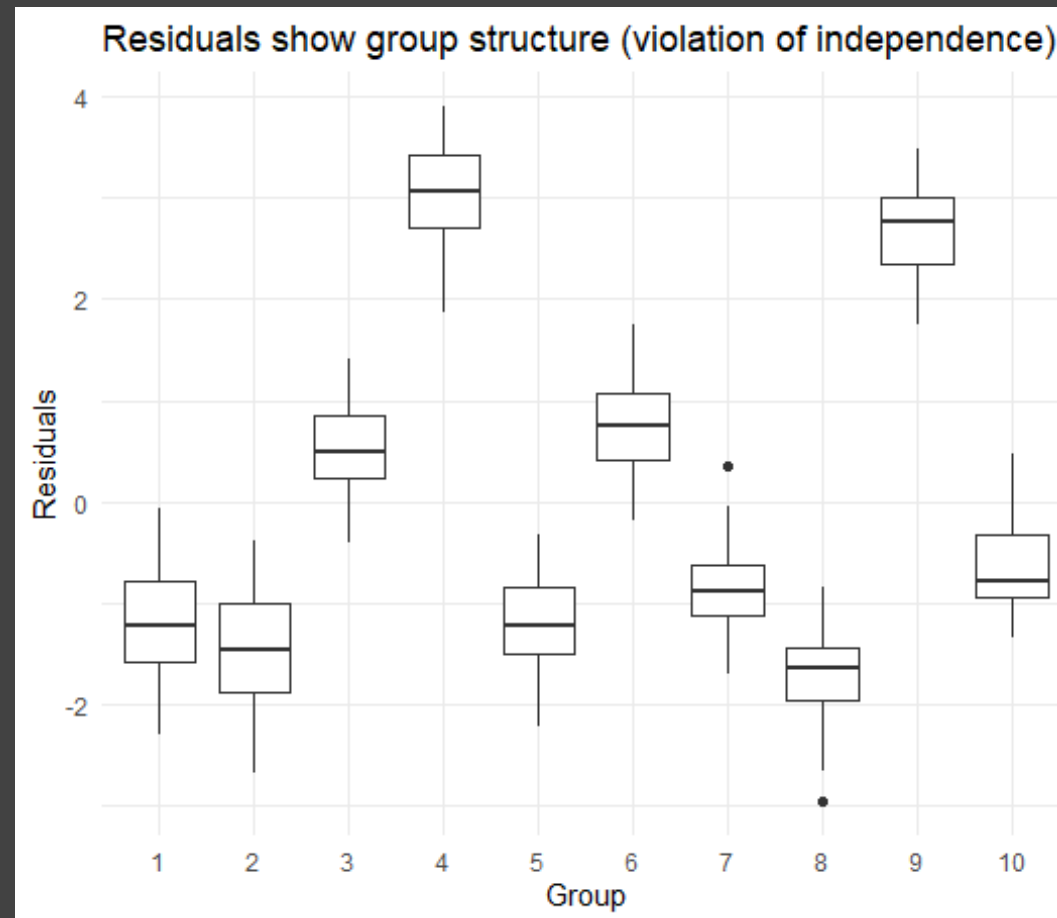
👉 **Observations are independent**

👉 **Residuals show no pattern**

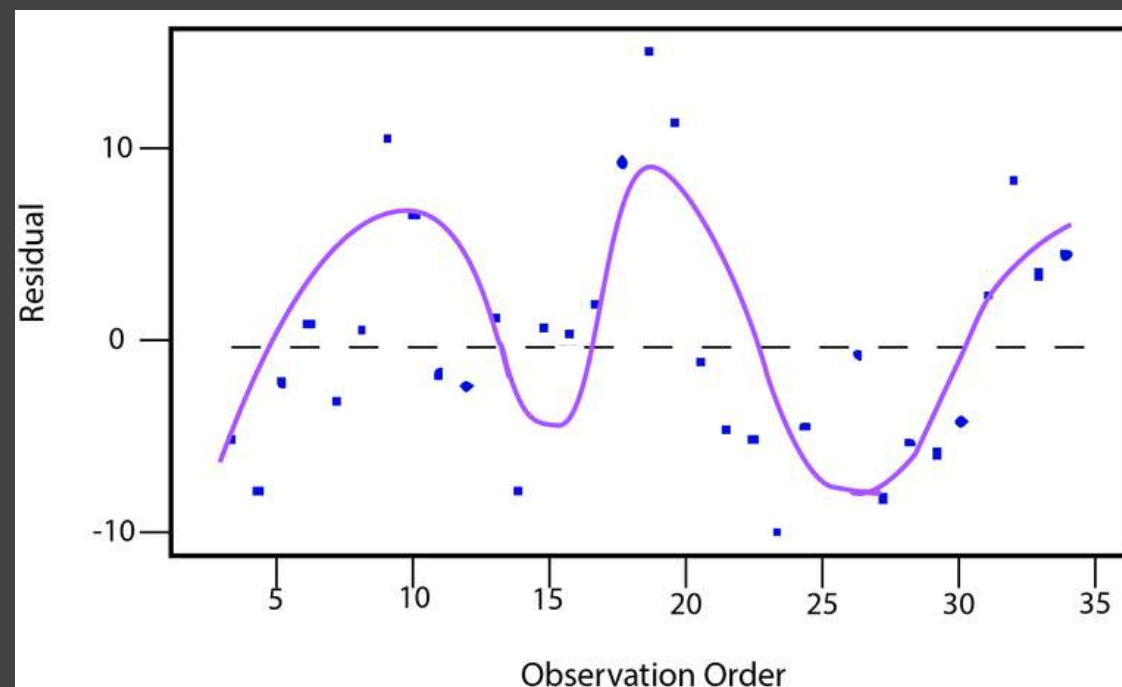
But some data are grouped or correlated:

- **Medical trials → patients within hospitals**
- **Spatial data → nearby locations are similar**
- **Time series data → autocorrelation**
- **Engineering data → repeated measures / sites**

What happens if we ignore this?



- Residuals show patterns
- Standard errors are wrong
- Inference becomes unreliable



Modeling Group Structure

◆ Fixed Effects

Estimate **specific differences** between **observed** groups

☞ Use when:

- Interested in each level explicitly
- Small, meaningful categories

◆ Random Effects

Model **variation** across groups

☞ Use when:

- Data has grouping / clustering structure
- Want to generalize **beyond observed** levels
- Many groups / repeated measures

⚠ Choose Random Effects over Fixed Effects

- Residual patterns by group → clustering, non-random structure
- Repeated measurements within groups → same subject, site, year
- Hierarchical data structure → observations nested in groups
- **Correlation within levels → observations not independent**

🔄 Same Variable, Different Modeling Choices

Year as fixed effect

☞ compare specific years → dummy variable

Year as random effect

☞ treat years as sampled → estimate variability/correlation across years → time series models

Space (*lon, lat*)

☞ GAM: smooth function $f(lon, lat)$

☞ GLMM: spatial correlation

☞ Choice depends on the question, not the variable itself

Advantages of Random Effects:

Handle correlation within groups

☞ Accounts for non-independence (core reason)

More efficient with many groups

☞ Avoid estimating a parameter for each level

Enable generalization

☞ Inference extends beyond observed groups

Partial pooling (shrinkage)

☞ Stabilizes estimates, especially with small sample sizes

Generalized Linear Mixed Model (GLMM)

$$g(y) = X\alpha + Z\beta + \varepsilon, y \sim E(\boldsymbol{\theta}), \beta \sim \Omega$$

What each part means

y → response variable

☞ Response follows an exponential family distribution
(Normal, Binomial, Poisson, etc.)

g(.) → link function
(connects mean of y to predictors)

Xα → fixed effects
(population-level effects, like LM/GLM)

Zβ → random effects
(group-level variation, e.g., site, patient)
☞ Random effects follow a distribution
(typically Normal with mean 0)

ε → residual error

Key Assumptions

- Observations are independent conditional on random effects
- Random effects are normally distributed
- Correct link function & response distribution
- Model captures group structure / correlation

What to say:

- GLMM extends GLM by adding random effects
- It allows us to model grouped or correlated data
- Independence is not removed—it's handled through random effects

☞ GLMM = GLM + Random Effects

Generalized Additive Mixed Model (GAMM)

$$g(y) = a + f(x) + Z\beta + \varepsilon, y \sim E(\boldsymbol{\theta}), \beta \sim \Omega$$

What GAMM adds

- Captures nonlinear relationships
- Keeps group structure (random effects)
- Combines flexibility + hierarchy

Practical Considerations

More flexible → higher risk of **overfitting**

Requires careful:

- smoothness selection
- diagnostics

Interpretation becomes less straightforward

👉 With flexibility comes responsibility

GAMM opens the door to advanced models:

- Spatiotemporal models
 - 👉 spatial + temporal smooths
- State-space models
 - 👉 latent processes over time
- Time-series models
 - 👉 autocorrelation + trends

 **Spatiotemporal , State-space, Time-series models can often be formulated within the GLMM/GAMM framework.**

 **If you are interested in learning these models, let us know, I'll be willing to teach**

Random Effects Structure in GLMM

Structure	What it assumes	When to use	R formula (glmmTMB)
Random Intercept + Slope	Groups differ in baseline <i>and</i> trend, and they are related	Most flexible / default assumption	glmmTMB(y ~ x + (x group), data = df)
Random Intercept only	Groups differ in baseline but are the same in trend	Groups start differently, but respond to x in the same way	glmmTMB(y ~ x + (1 group), data = df)
Random Slope only	Same baseline across groups, but different trends	When groups start similarly but evolve differently	glmmTMB(y ~ x + (0 + x group), data = df)

Random Effects Examples

🧠 1. Random Intercept Only

📌 Example: Hospital patient recovery

y: recovery time

X: treatment dose

Group: hospital

👉 Interpretation:

Hospitals differ in baseline recovery
(some are generally faster)

BUT treatment effect is the same
everywhere

↗ R

```
library(glmmTMB)
```

```
model_ri <- glmmTMB(  
  recovery ~ dose + (1 | hospital),  
  data = df  
)
```

🧠 2. Random Slope Only

📌 Example: Fertilizer effect in identical
greenhouses

y: plant growth

X: fertilizer amount

Group: greenhouse

👉 Interpretation:

All greenhouses start at same baseline
BUT respond differently to fertilizer

↗ R

```
model_rs <- glmmTMB(  
  growth ~ fertilizer + (0 + fertilizer | greenhouse),  
  data = df  
)
```

🧠 3. Random Intercept + Slope

📌 Example: Salamander abundance in
ponds

y: salamander count

X: temperature

Group: pond / site

👉 Interpretation:

Some ponds have more salamanders
overall

AND temperature effect differs by pond

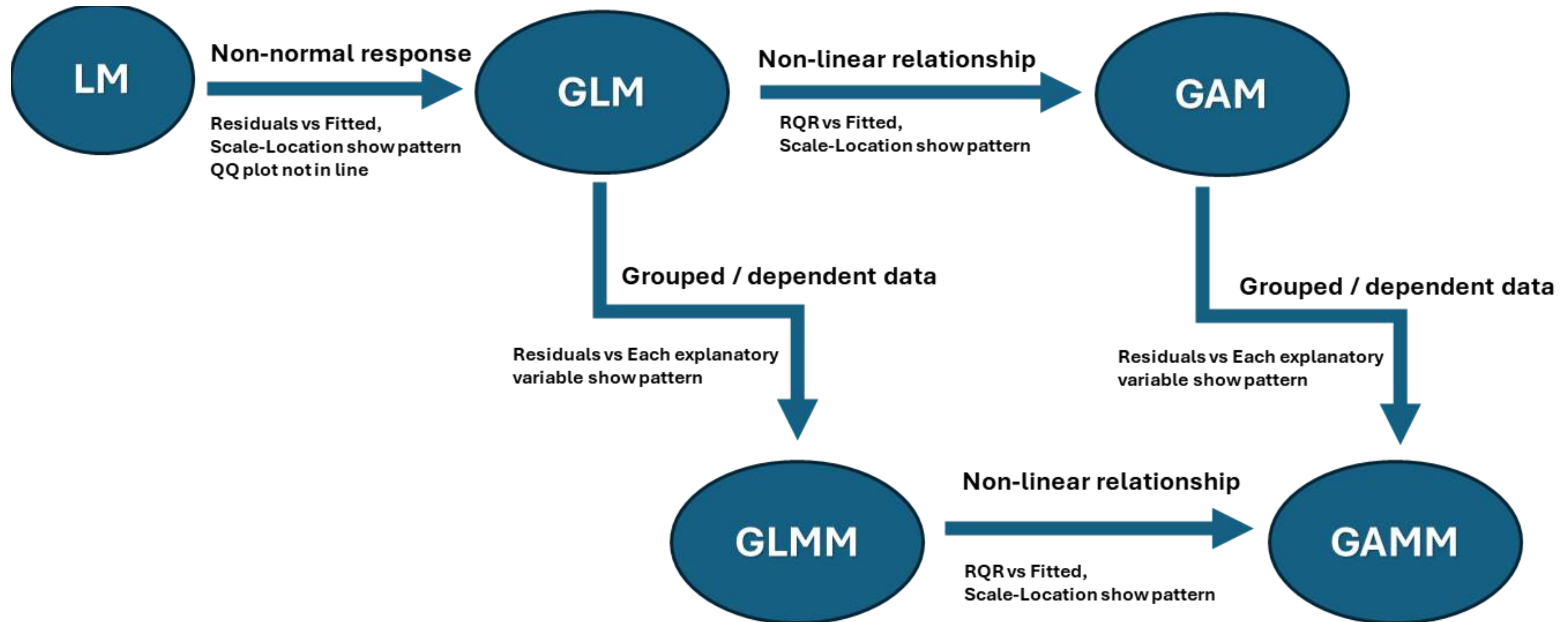
↗ R

```
model_ris <- glmmTMB(  
  count ~ temperature + (temperature | pond),  
  data = df  
)
```

R Packages for GLMM / GAMM

Package	Main Use Case	Model Function	Prediction	Residuals	Optimization / Engine	What to Check After Fit
lme4	Classic GLMMs (no spatial structure)	lmer(), glmer()	predict()	residuals()	Laplace / (P)QL	Residual vs fitted, QQ plot
glmmTMB	Flexible GLMMs (Poisson, NB, zero-inflation)	glmmTMB()	predict()	residuals()	TMB (fast C++ optimization)	RQR / DHARMA diagnostics
mgcv	GAM / GAMM (smooth effects)	gam(), gamm()	predict.gam()	residuals()	Penalized likelihood (REML/GCV)	Smooth plots, k-index, residuals
sdmTMB	Large-scale spatial-temporal	sdmTMB()	predict()	residuals()	SPDE / GMRF (TMB backend)	Spatial sd error maps
staRve	Large-scale spatial-temporal	strv_fit()	strv_predict()	residuals()	NNGP (TMB backend)	Spatial sd error maps
momentuHMM	Animal movement / state-switching models	fitHMM()	predict()	pseudo-residuals	EM algorithm	State decoding, residual autocorrelation

Evolution of Statistical Models



PART 2

Applications

May 26th, 2026

About Me

- **MSc in Statistics, Dal**
- **Data quality control analyst, Dal**
- **Statistical modeling specialist**
- **Workshop facilitator**
- **New ocean diver**



LinkedIn



github



Workshop Overview

- 03** Review statistical models
- 04** Introduction of R & R markdown
- 05** Output interpretations
- 06** Case studies
- 07** Future workshops

Model Review

Model	When to Use	Key Assumptions	R Package / Function	Residual Diagnostics to Check
LM (Linear Model)	Response is continuous and relationships are approximately linear	Linear relationship, independent observations, constant variance (homoscedasticity), normally distributed residuals	stats::lm()	Residual vs fitted plot (linearity, variance), Q-Q plot (normality), leverage/outliers, VIF
GLM (Generalized Linear Model)	Non-normal response (counts, proportions, positive continuous data), known link function	Correct distribution family (Gaussian, Poisson, Binomial, Gamma, etc.), correct link function, independence of observations	stats::glm()	Residual patterns, Q-Q plot (randomized quantile residuals), overdispersion, zero inflation, VIF
GAM (Generalized Additive Model)	Relationship is nonlinear but observations are independent	Correct distribution/link, smooth functions adequately capture nonlinear patterns, independence	mgcv::gam()	Residual plots, Q-Q plot, basis dimension check (gam.check()), concurvity, overdispersion
GLMM (Generalized Linear Mixed Model)	Hierarchical/grouped data (e.g., students within schools, fish within sites, penguins within species)	GLM assumptions + normally distributed random effects, correct grouping structure	glmmTMB::glmmTMB(), lme4::glmer()	Same as GLM Also need to check normality assumption for random effects
GAMM (Generalized Additive Mixed Model)	Nonlinear relationships and grouped/hierarchical data	GAM assumptions + normally distributed random effects, appropriate smoothness and grouping structure	mgcv::gamm(), mgcv::gamm4(), glmmTMB with smooth terms	Same as GAM Also need to check normality assumption for random effects

Why R Still Matters?

Advantages of R

● R = NOT JUST ANOTHER LANGUAGE

Built for statistics first, not added later

Forces better habits:

- model thinking
- assumptions
- interpretation

● WHERE R STANDS OUT

- Research & academia
- Government & science (e.g., fisheries, health)
- Reproducible analysis
- Statistical model packages

💡 Key idea (say this out loud):

“Python helps you build systems. R helps you understand data.”

👉 If your goal is insight, not just code, R is powerful.

● R MARKDOWN

A framework to combine:

- text (your explanation)
- code (your analysis)
- output (plots, tables)

👉 Output formats:

- HTML
- PDF
- Word
- Slides

💡 Think:

Jupyter Notebook → flexible exploration

R Markdown → structured, publishable analysis

It's always good to know more more programming languages 😊

Key R Libraries

Library	What it is	When to use
ggplot2	Visualization system (grammar of graphics)	When you want clean, consistent, publication-quality plots
dplyr	Data manipulation (filter, select, mutate)	When cleaning and transforming datasets
tidyr/tidyverse	Data reshaping	When converting between wide/long formats
lme4 / glmmTMB	Mixed-effects models	When data has grouping (e.g., repeated measures)
mgcv	GAM models	When relationships are nonlinear
knitr	Engine behind R Markdown	When generating reports
rmarkdown / quarto	Reporting framework	When creating reproducible documents

Important Code Structures

1. Data wrangling

```
# Example: Converting wide data to long format
long_data <- wide_data %>%
  pivot_longer(
    cols = c("Jan", "Feb", "Mar"), # The columns to gather
    names_to = "Month",            # New column name for original headers
    values_to = "Sales"           # New column name for the values
  )
```

- ***filter()***: Subsets rows based on specific logical criteria.
- ***select()***: Extracts specific columns by name.
- ***mutate()***: Creates new variables or modifies existing ones.
- ***group_by()***: Groups data into subsets for collective operations.
- ***summarize()***: Reduces multiple values down to a single summary statistic.
- ***arrange()***: Reorders rows based on the values of one or more variables

2. Modeling

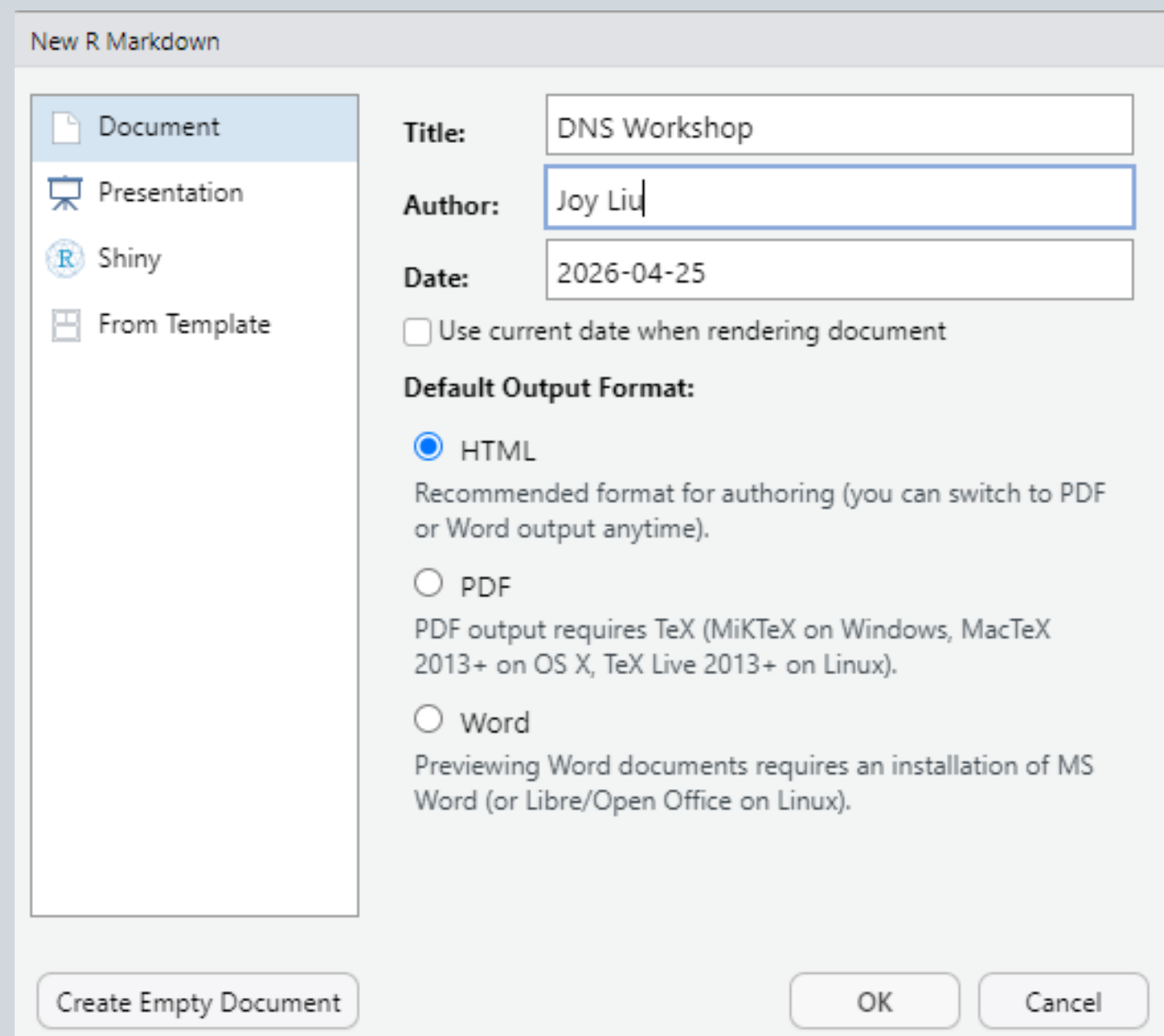
```
model <- lm(y ~ x1 + x2, data = df)
summary(model)
```

3. Visualization

```
ggplot(data = mpg, mapping = aes(x = displ, y = hwy, color = class)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Engine Size vs. Fuel Efficiency",
        x = "Engine Displacement (L)",
        y = "Highway MPG") +
  theme_minimal()
```

- ***ggplot()***: The initialization function that creates the "blank canvas" for your plot. It typically takes the data as its first argument.
- ***aes()***: Stands for Aesthetics. This function maps your data's variables to visual properties, such as the x and y axes, color, size, or shape.
- ***geom_point()***: A geometric object layer that adds individual data points to create a scatter plot.
- ***geom_smooth()***: Adds a statistical summary layer, such as a regression line, to help identify trends in the data.
- ***labs()***: Used to modify labels like the plot title and axis names.
- ***theme_minimal()***: One of many pre-defined themes that change the overall appearance (e.g., background, grid lines, and fonts) to be cleaner.

R Markdown



The 'New R Markdown' dialog box is shown. On the left, there is a sidebar with four options: 'Document' (selected), 'Presentation', 'Shiny', and 'From Template'. The main area contains fields for 'Title' (DNS Workshop), 'Author' (Joy Liu), and 'Date' (2026-04-25). Below these fields is a checkbox 'Use current date when rendering document' which is unchecked. Under 'Default Output Format', there are three radio buttons: 'HTML' (selected), 'PDF', and 'Word'. Descriptive text is provided for each format: HTML is recommended for authoring; PDF requires TeX; Word requires MS Word or Libre/Open Office. At the bottom, there are three buttons: 'Create Empty Document', 'OK', and 'Cancel'.

Title: DNS Workshop

Author: Joy Liu

Date: 2026-04-25

☐ Use current date when rendering document

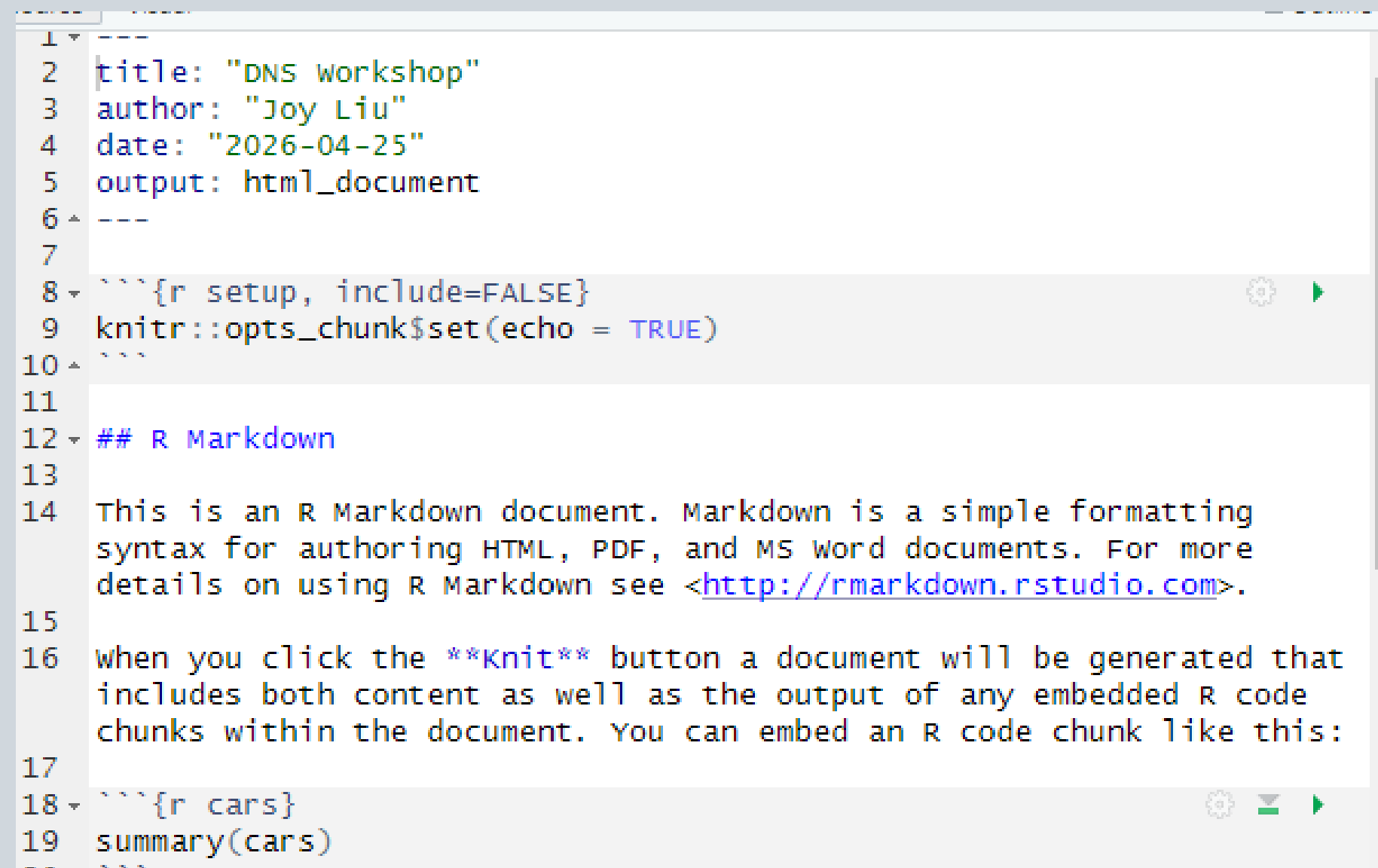
Default Output Format:

☒ HTML
Recommended format for authoring (you can switch to PDF or Word output anytime).

☐ PDF
PDF output requires TeX (MiKTeX on Windows, MacTeX 2013+ on OS X, TeX Live 2013+ on Linux).

☐ Word
Previewing Word documents requires an installation of MS Word (or Libre/Open Office on Linux).

Create Empty Document OK Cancel



The R Markdown code editor shows a document with the following content:

```
1 ---
2 title: "DNS workshop"
3 author: "Joy Liu"
4 date: "2026-04-25"
5 output: html_document
6 ---
7
8 ```{r setup, include=FALSE}
9 knitr::opts_chunk$set(echo = TRUE)
10 ```
11
12 ## R Markdown
13
14 This is an R Markdown document. Markdown is a simple formatting
15 syntax for authoring HTML, PDF, and MS word documents. For more
16 details on using R Markdown see <http://rmarkdown.rstudio.com>.
17
18 when you click the **knit** button a document will be generated that
19 includes both content as well as the output of any embedded R code
20 chunks within the document. You can embed an R code chunk like this:
21
22 ```{r cars}
23 summary(cars)
24 ```
```

- **Markdown Formatting:** # Header, ## Subheader, bold, or italics are used to format text, just like a standard markdown file.
- **Inline Code:** Using ````{r, ...}```` to create a chunk, you can run a quick R calculation inside a sentence.
- **Common Chunk Options:**
 1. `echo = TRUE` : The code and its results are both shown
 2. `eval = FALSE`: Displays the code, but does not run it.
 3. `message = FALSE` / `warning = FALSE`: Hides messages or warnings from R packages.

How to Write a Clear Statistical Report

1. Introduction

Background: What's the context?

💡 Example:

- Sea scallops (*Placopecten magellanicus*) are a major fishery in Atlantic Canada
- Scientists estimate biomass to support decisions by Fisheries and Oceans Canada
- Meat weight is harder to measure than shell height

Research question(s): What are we trying to answer?

💡 Example:

- Can we improve estimation of scallop meat weight from shell height?
- How do spatial and temporal factors affect this relationship?

Why it matters?

💡 Example:

- Accurate estimates → better fishery management
- Poor models → risk of over/under harvesting

Brief idea of methods

💡 Example:

- spatiotemporal modeling (STM)
- 10-fold spatial cross-validation

2. Exploratory Data Analysis (EDA)

Understand before modeling

- Data source + variables (what each represents)
- Data cleaning: missing values, outliers, transformations

Identify:

- response variable
- predictors
- Key visuals:
 1. Distribution (histogram / density)
 2. Correlation (check multicollinearity)
 3. Scatter plots (relationships)
 4. Boxplots / bar plots (group differences)
 5. Maps (if spatial)

How to Write a Clear Statistical Report

3. Method

- Problem–Method Alignment: What challenges does the data have?

👉 Example:

“Scallop meat weight varies across space and time → simple models are not sufficient”

- Model Choice: Introduce the model in plain language
 1. What is it?
 2. When do we use it?
 3. What problem does it solve?
- Mathematical Formulation

4. Results

- Model diagnostics:
 1. residual plots
 2. goodness-of-fit
 3. Model comparison:
 4. AIC
 5. k-fold cross-validation
- Final model selection
- Interpretation:
 1. What do coefficients mean?
 2. Are effects significant?
 3. Prediction (if relevant)
 4. Standard errors

5. Conclusion

- Summarize key findings
- Answer the research question
- Highlight limitations

6. Future Work / Discussion

- What could be improved?
- What’s missing?
- Next steps
- 💡 Often combined as “Discussion”

Model Interpretations-LM

```
Call:
lm(formula = y ~ x)

Residuals:
    Min       1Q   Median       3Q      Max
-41.925 -15.682  -2.145   10.562  120.855

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -116.3716    15.2438  -7.634  2.4e-12 ***
x             2.0187     0.1943  10.389  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 22.67 on 151 degrees of freedom
Multiple R-squared:  0.4168,    Adjusted R-squared:  0.413
F-statistic: 107.9 on 1 and 151 DF,  p-value: < 2.2e-16
```

From the output:

Intercept estimate = -116.37

x estimate = 2.02

x significance = *** ($p < 0.001$)

This suggests that x is a significant explanatory variable for predicting y.

Interpretation:

The intercept represents the expected value of y when $x = 0$.

The slope means that for every 1-unit increase in x, the predicted value of y increases by approximately 2.02 units, **assuming all else remains constant.**

LM Output Component	What It Is	What It Tells Us
Estimate	The estimated regression coefficient for each predictor	The expected change in the response variable for a one-unit increase in the predictor (holding other variables constant). Positive = increase; negative = decrease.
Std. Error	Standard error of the coefficient estimate	Measures uncertainty/precision of the estimate. Smaller values indicate more reliable coefficient estimates.
t value	Test statistic (Estimate / Std. Error)	Indicates how far the estimate is from zero relative to its uncertainty. Larger absolute values suggest stronger evidence that the predictor has a real effect.
Pr(> t)	p-value for hypothesis test of coefficient = 0	Shows whether the predictor is statistically significant. Small p-values (typically < 0.05) suggest the predictor contributes meaningfully to the model.
Significance Codes	Symbolic shorthand for p-values (***, **, *, .)	Quick visual summary of predictor significance. More stars = stronger statistical evidence.
Residual Standard Error (RSE)	Standard deviation of model residuals	Measures average prediction error in the response variable's units. Lower RSE indicates better fit.
Degrees of Freedom (DF)	Sample size minus number of estimated parameters	Reflects how much information is available for estimating variability and testing significance.
Multiple R-squared	Proportion of variance explained by the model	Indicates overall model fit. Higher values mean the model explains more variation in the response.
Adjusted R-squared	R-squared adjusted for number of predictors	Penalizes unnecessary predictors. Better for comparing models with different numbers of variables.
F-statistic	Overall significance test for the full model	Tests whether the model explains significantly more variation than a null model with no predictors. Large F + small p-value = model is useful overall.
F-statistic p-value	p-value for overall model test	Indicates whether at least one predictor contributes significantly to explaining the response.

Model Interpretations-GLM

Distribution	Common Link	Model Form	Coefficient Interpretation	Back-transformed Meaning
Gaussian	Identity	$\mu = X\beta$	1-unit increase in X changes Y by β units	Direct effect
Poisson	Log	$\log(\mu) = X\beta$	1-unit increase in X changes log count by β	$\exp(\beta)$ = multiplicative change in expected count
Binomial	Logit	$\text{logit}(p) = X\beta$	1-unit increase in X changes log-odds by β	$\exp(\beta)$ = odds ratio
Gamma	Log	$\log(\mu) = X\beta$	1-unit increase in X changes log mean by β	$\exp(\beta)$ = proportional change in mean

Model Interpretations-GLMM

What to Check	How to Check		Good Result	Warning Sign	Meaning
Normality of random effects	<code>check_model(model)</code> Q-Q plot		Dots follow diagonal line	Curvature, S-shape, outliers	Random effects approximately normal
Group variance size	<code>summary(model)</code> or <code>VarCorr(model)</code>		Moderate/meaningful variance	Near zero variance	Whether grouping structure matters
Overdispersion	<code>check_overdispersion(model)</code>		Ratio ~1	Large ratio	Distribution may be misfit
Singular fit	<code>check_singularity(model)</code>		FALSE	TRUE	Random structure may be too complex
Residual diagnostics	<code>check_model(model)</code>		Uniform residuals	Tail deviations	Distribution/link may need revision

Note: Check if random effects are significant

Case Study

Objective: Predict penguin body mass (g) using morphological and demographic characteristics

Study Location: Three islands in the Palmer Archipelago, Antarctica

- Biscoe Island
- Dream Island
- Torgersen Island

Study Period: 2007–2009

Species:

- Adelie Penguin
- Chinstrap Penguin
- Gentoo Penguin

Response Variable: Body mass (g)

Predictors: Bill length, bill depth, flipper length, sex, island, year, and species

Future workshops

Statistical Modeling in R Workshop
Series



Thank You



<https://www.linkedin.com/in/joy-liu-msc-statistics-013121196/>



<https://github.com/joyliujoyliu>



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